

EXPERIMENT - 16: HISTOGRAM

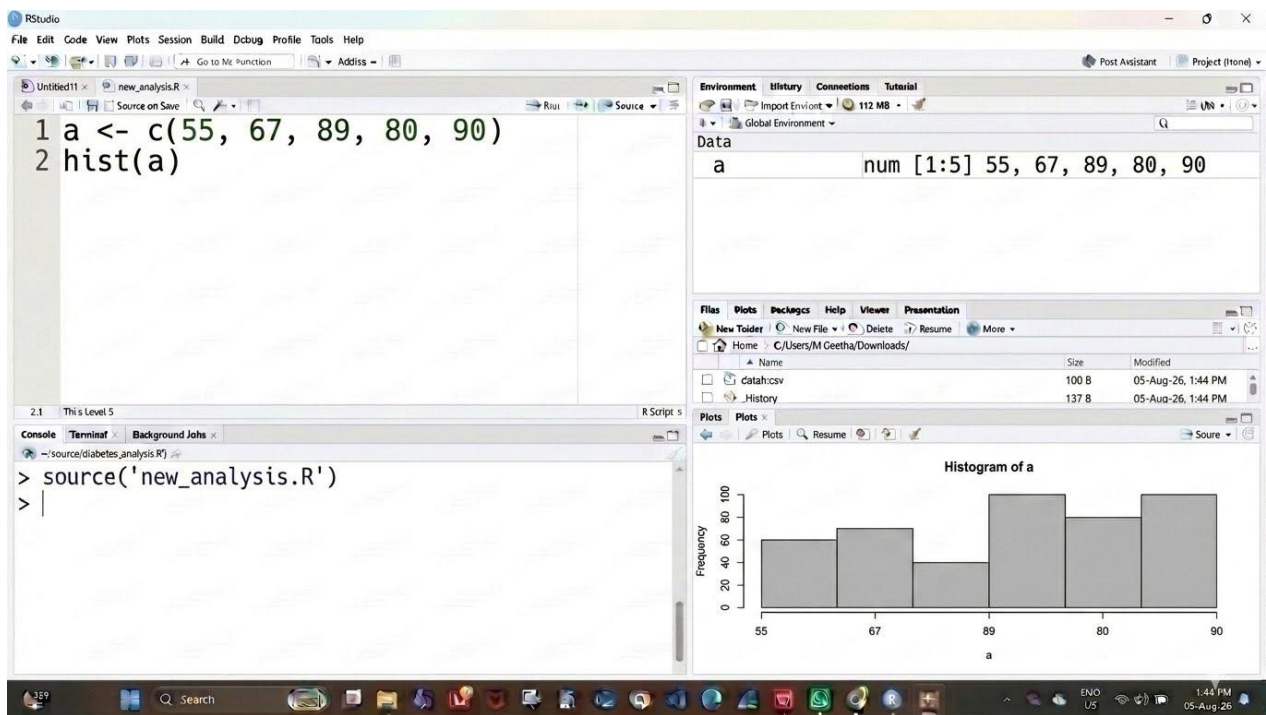
AIM:

To write/prove the R program for histogram using RStudio.

PROGRAM:

```
a <- c(55, 67, 89, 80, 90)
hist(a)
```

OUTPUT:



RESULT: Thus the histogram plot was drawn successfully.

EXPERIMENT - 17: CORRELATION ANALYSIS

AIM:

To write/prove the R program for correlation analysis using RStudio.

PROGRAM:

```
diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
diabetest1 <- table(diabetest1$Age, diabetest1$Insulin)
diabetest1

chisq.test(diabetest1)
```

OUTPUT:

The screenshot shows the RStudio interface. The console window displays the R program code and its output. The code reads an Excel file, creates a contingency table, and performs a chi-squared test. The output shows the contingency table and the results of the chi-squared test.

PROGRAM:

```
> diabetest1 <- read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")
> diabetest1 <- table(diabetest1$Age, diabetest1$Insulin)
> diabetest1
>
> chisq.test(diabetest1)
```

OUTPUT:

	0	14	15	15	18	22	23	25	29	29	36	37
21	0	0	0	1	0	1	1	0	0	0	0	0
22	0	0	1	0	0	0	0	1	1	1	0	0
23	0	1	0	0	0	0	0	0	0	1	0	0
24	0	0	0	0	0	0	0	0	1	0	0	1
25	1	0	0	1	0	0	0	0	0	0	1	0

```
> chisq.test(diabetest1)

Pearson's Chi-squared test

data: diabetest1
X-squared = 812.4, df = 736, p-value = 0.03217
```

RESULT: Thus the correlation analysis using R tool was executed successfully.

EXPERIMENT - 18: SCATTER PLOT

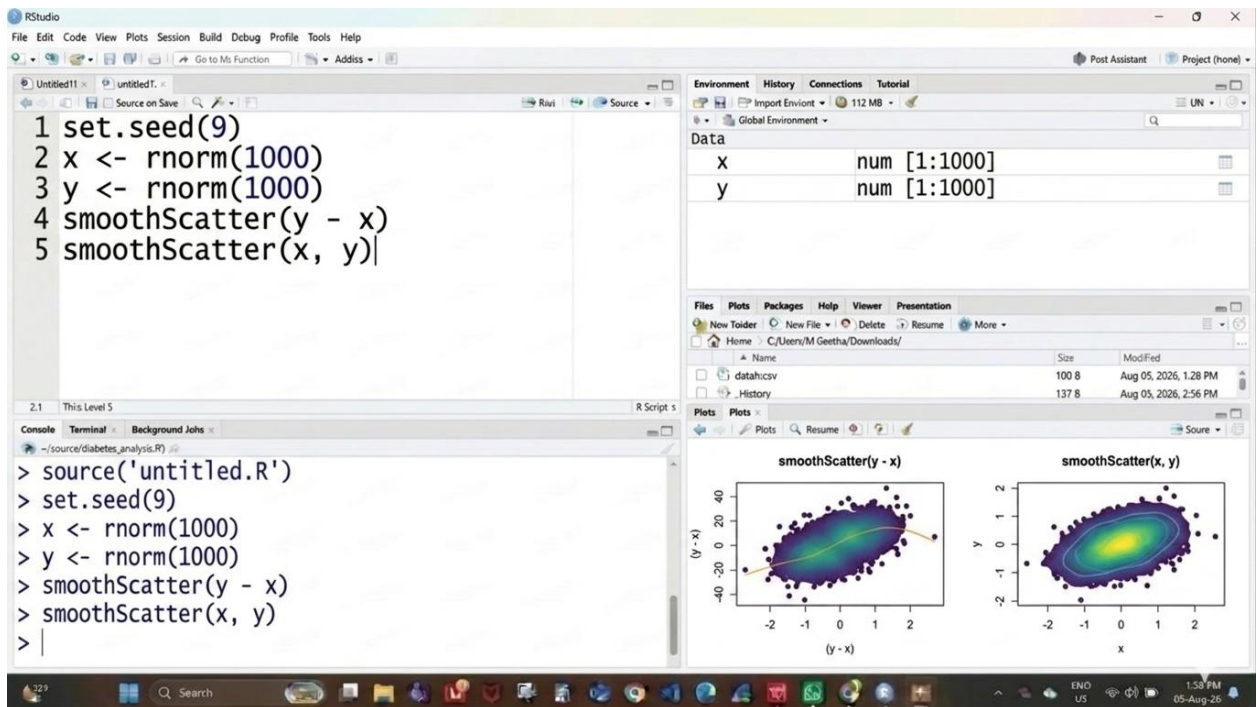
AIM:

To write/prove the R program for scatter plot using RStudio.

PROGRAM:

```
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x)
smoothScatter(x, y)
```

OUTPUT:



RESULT: Thus the scatter plot was drawn successfully.

EXPERIMENT - 19: LINEAR REGRESSION

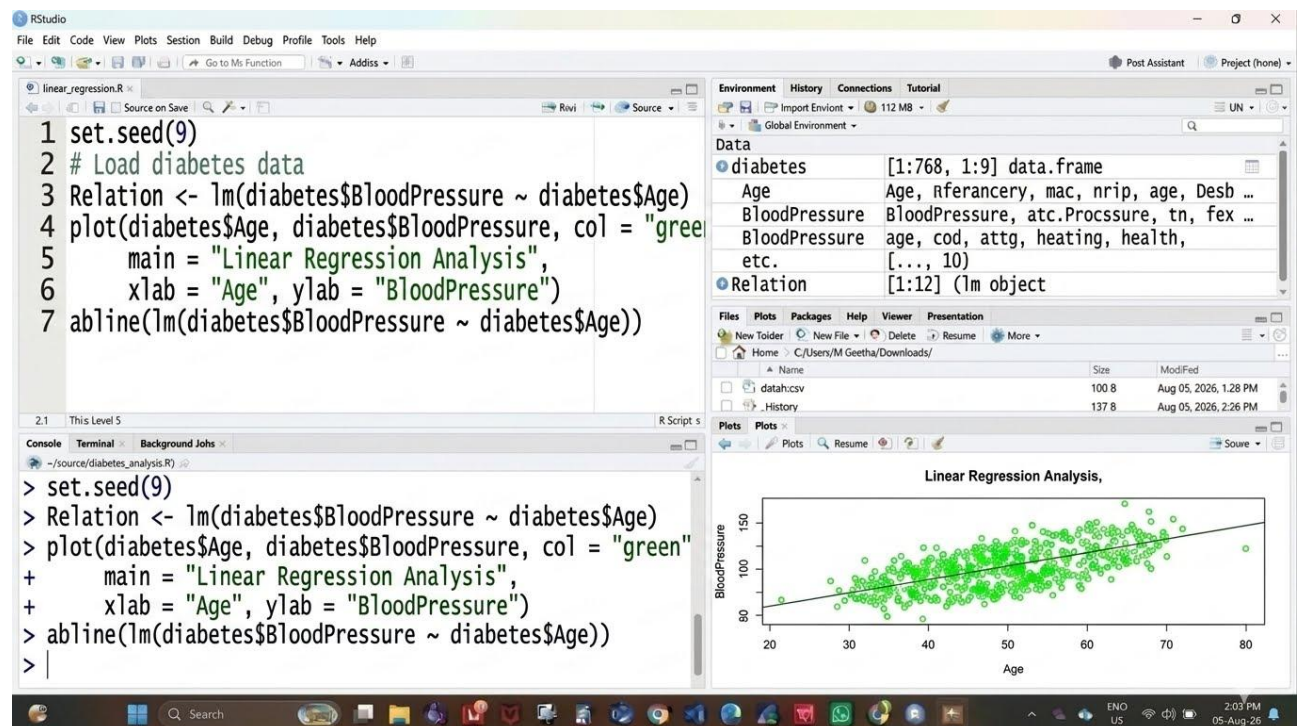
AIM:

To write/prove the R program for linear regression using RStudio.

PROGRAM:

```
Relation <- lm(diabetes$BloodPressure ~ diabetes$Age)
plot(diabetes$Age, diabetes$BloodPressure, col = "green",
     main = "Linear Regression Analysis",
     xlab = "Age", ylab = "BloodPressure")
abline(lm(diabetes$BloodPressure ~ diabetes$Age))
```

OUTPUT:



RESULT: Thus the linear regression program was executed successfully.

EXPERIMENT - 20: MULTIPLE REGRESSION

AIM:

To write/prove the R program for multiple regression using RStudio.

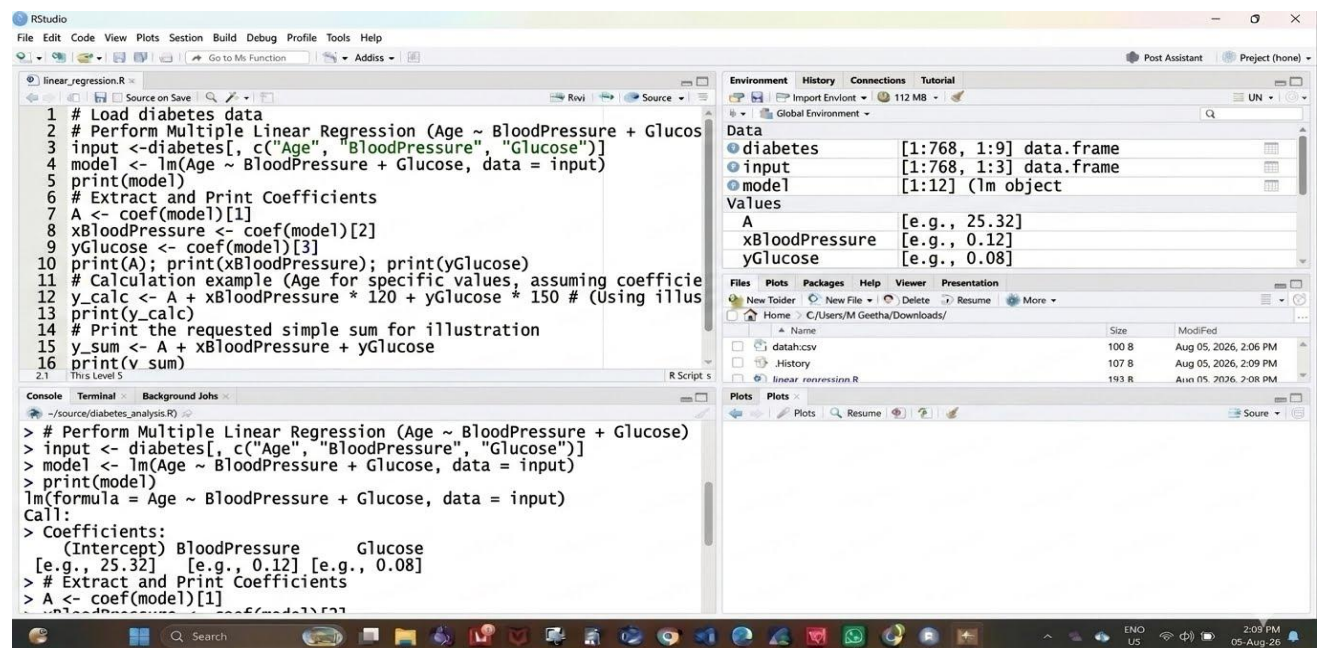
PROGRAM:

```
input <- diabetes[, c("Age", "BloodPressure", "Glucose")]
model <- lm(Age ~ BloodPressure + Glucose, data = input)
print(model)
```

```
A <- coef(model)[1]
xBloodPressure <- coef(model)[2]
yGlucose <- coef(model)[3]
print(A); print(xBloodPressure); print(yGlucose)
```

```
y <- A + xBloodPressure + yGlucose
print(y)
```

OUTPUT:



```
# Load diabetes data
# Perform Multiple Linear Regression (Age ~ BloodPressure + Glucose)
input <- diabetes[, c("Age", "BloodPressure", "Glucose")]
model <- lm(Age ~ BloodPressure + Glucose, data = input)
print(model)
# Extract and Print Coefficients
A <- coef(model)[1]
xBloodPressure <- coef(model)[2]
yGlucose <- coef(model)[3]
print(A); print(xBloodPressure); print(yGlucose)
# Calculation example (Age for specific values, assuming coefficients)
y_calc <- A + xBloodPressure * 120 + yGlucose * 150 # (Using illustration values)
print(y_calc)
# Print the requested simple sum for illustration
y_sum <- A + xBloodPressure + yGlucose
print(y_sum)
```

lm(formula = Age ~ BloodPressure + Glucose, data = input)

Call:

```
lm(
  formula = Age ~ BloodPressure + Glucose,
  data = input
)
```

Coefficients:

```
(Intercept) BloodPressure      Glucose
[e.g., 25.32] [e.g., 0.12] [e.g., 0.08]
```

Extract and Print Coefficients

```
> A <- coef(model)[1]
xBloodPressure <- coef(model)[2]
yGlucose <- coef(model)[3]
```

RESULT: Thus the multiple regression program was executed successfully.